



DuoMix Fader

Play two apps at once, mix them with a crossfader — and see, while the music plays, which scale it is in and which chords it loops through.

User guide · version 0.5 · September 2026

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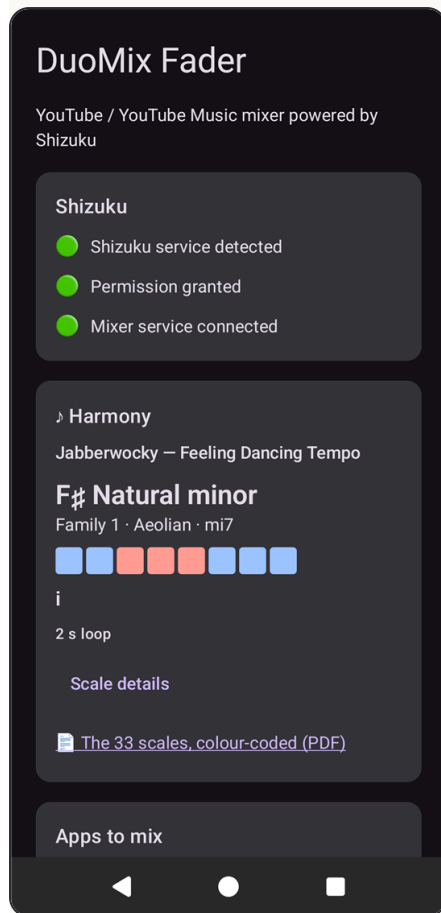
1 • What it does

Android normally lets one app play at a time: start a video and your music pauses. DuoMix Fader removes that limit for two apps you choose — a **music** app and a **video or voice** app — and gives you a two-channel mixer with a DJ-style, constant-power crossfader. It needs no root.

On top of the mixer, the **Harmony** feature listens to the music channel and estimates the scale of the piece among the 7 families and 33 *Scales of Harmonies* described by Oliver Prehn (NewJazz), its tonic, and its chord progression in Roman numerals. It is an accompaniment aid: a steady, averaged reading of the piece, not a note-by-note transcription.

DuoMix Fader is an independent project. It is not affiliated with Google, YouTube, the Shizuku project or Oliver Prehn.

2 • Getting started



Three green lights: Shizuku is running, permission is granted, the mixer service is connected.

DuoMix Fader relies on **Shizuku**, a separate free app that lets it perform a few system-level audio operations without root.

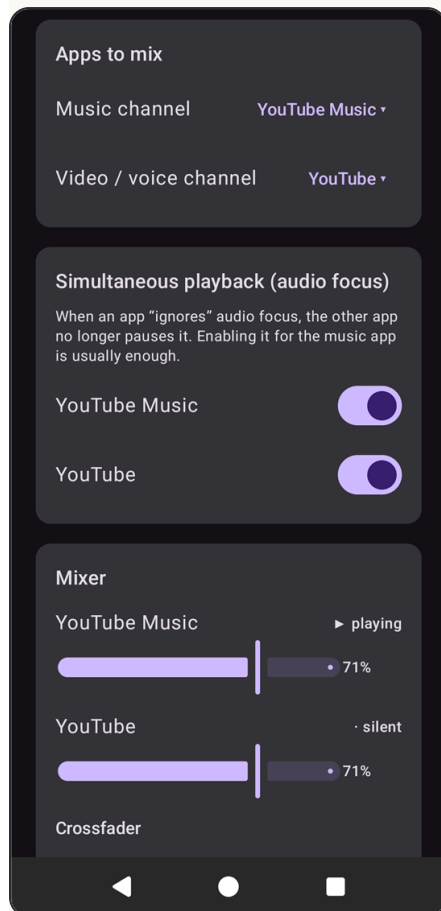
1. Install **Shizuku** and start it through *Developer options* → *Wireless debugging*, following Shizuku's own instructions. After a reboot it has to be started again, unless you enabled its start-on-boot option.
2. Open **DuoMix Fader** and tap **Request Shizuku permission**, then accept.
3. When the three lights are green, you are ready.

If the first light is red, Shizuku is not running: open Shizuku and start it. Nothing in DuoMix Fader works without it.

Languages

The app follows your phone's language — English, French or Spanish. You can also choose its language alone in *Settings* → *System* → *Languages* → *App languages*. Scale names are international musical terms and are never translated.

3 • The mixer



Apps to mix, the audio-focus switches, the two channel sliders and the crossfader.

Apps to mix

Pick one app per channel. Only supported apps that are installed are listed: for music, YouTube Music, Spotify, Deezer, Apple Music, Amazon Music, SoundCloud, Tidal, Qobuz — and YouTube, X and Telegram, since music is also played from videos and feeds (Harmony listens to the music channel); for video and voice, YouTube, X, Telegram, Twitch, VLC, Netflix, Pocket Casts, Chrome, Firefox. An app can hold only one channel: pick it on a channel while the other one has it, and the two channels swap. The list is closed on purpose: the privileged service refuses any other app.

Simultaneous playback

Android pauses an app when another one takes the *audio focus*. Turn the switch on for an app and it stops being paused by the other. Enabling it for the **music app** is usually enough.

Some apps pause *themselves* when they are refused audio focus (X does). Their switch is disabled and says so: enable the option on the other channel instead.

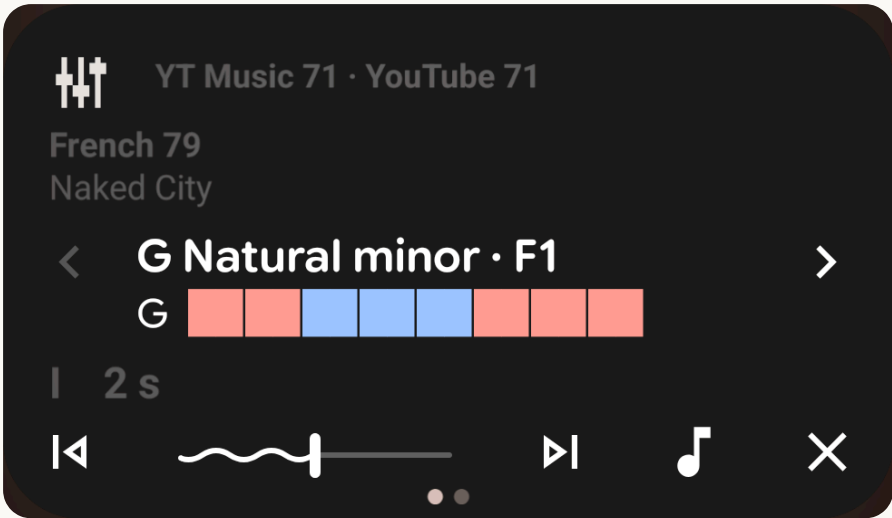
Volumes and crossfader

Each slider sets the volume of one app only. The crossfader moves from the music app (left) to the video app (right) following an equal-power curve: at the centre both sit at 71 %, so the overall loudness stays constant.

When you close both the screen and the notification, both apps get their full volume back, and switching a channel to another app restores the app you leave.

4 • The fader in your notifications

You do not have to keep the app open. While DuoMix Fader runs, a media-player card called **DuoMix fader** appears next to your other players — in the notification shade and on the lock screen.

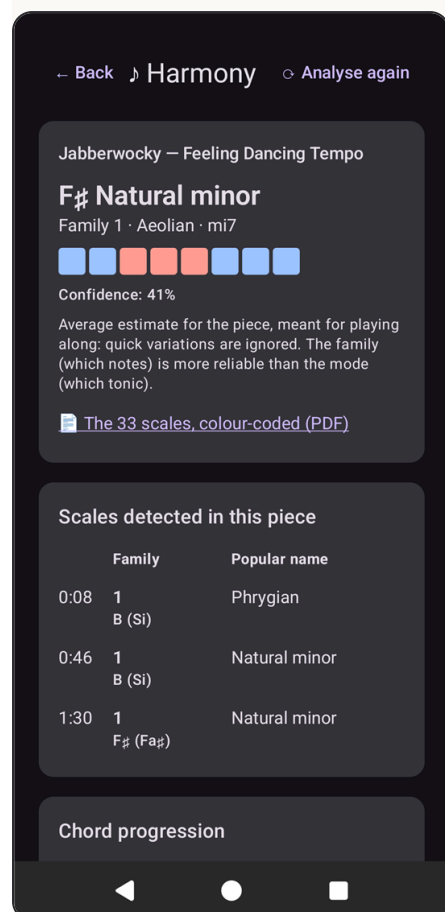


The card, cropped from the notification shade. Here: G natural minor, family 1 — G A in red, B $\flat$ C D in blue, E $\flat$ F G in red — a few seconds into a piece, when the progression is still a single chord. The name of the audio output is masked.

ON THE CARD	WHAT IT SHOWS OR DOES
Title	The essential: the mode. Tonic, popular name and family — “E Mixolydian · F1”.
Colour tiles	The key, in letter notation — “A”, “D $\flat$ ” — then one tile per note of the scale, from the tonic to its octave . Blue and red are the two whole-tone sets of the keyboard, so a change of colour means a half step or a step and a half. A minor reads R R B B B R R R. This sequence is the fingerprint of the mode.
Chord progression	Under the tiles, in Roman numerals relative to the tonic: “i – bVI – iv – bVI 8 s” when a loop was found, with its length; “... i – iv – bVII” — the last chords heard — when none was. It only changes once a new progression has held for four seconds.
Small text, top	To the right of the app icon, the two apps and their volumes: “YT Music 71 · YouTube 71”. Beneath, on two lines, the artist and the title of the piece; a long title is cut short. This text is drawn in the card’s background artwork, which Android darkens.
Progress bar	It is the crossfader: drag it. Left is the music app, right is the video app. To re-centre, drag it to the middle.
⏮ / ⏭	Move the fader 10 % towards music / towards video.
🎵	Close the shade and open the Harmony panel.
✕	Hide the card. It comes back the next time you open the app.

DuoMix Fader plays no sound itself: this player card is only a control surface, which is why it has no play button. Its layout, its two text lines and their size belong to Android. Nothing scrolls or blinks on the card: it only changes when the scale, the piece or the balance does.

5 • Harmony: scale, tonic, chords



The piece as announced by the music app, the scale found, the scales held during the piece, and the chord progression.

Open it from the **Harmony** card of the main screen (*Scale details*) or from the **Harmony** button of the notification. Analysis starts by itself as soon as the music app plays.

What you read

- **The scale** — tonic and popular name, then its family, systematic name and chords, as in Oliver Prehn's table. The tonic is written in your language's system, with the other one next to it: "B $\flat$ (Si $\flat$)".
- **Confidence** — how clearly this set of notes beats the next candidate. Below about 15 %, treat the family as a hint.
- **Scales detected in this piece** — one row per scale held, with the time it appeared and the five columns of the table. Scroll the table sideways.
- **Chord progression** — Roman numerals relative to the tonic, the chord names beneath, and the length of the loop.

Reading the numerals

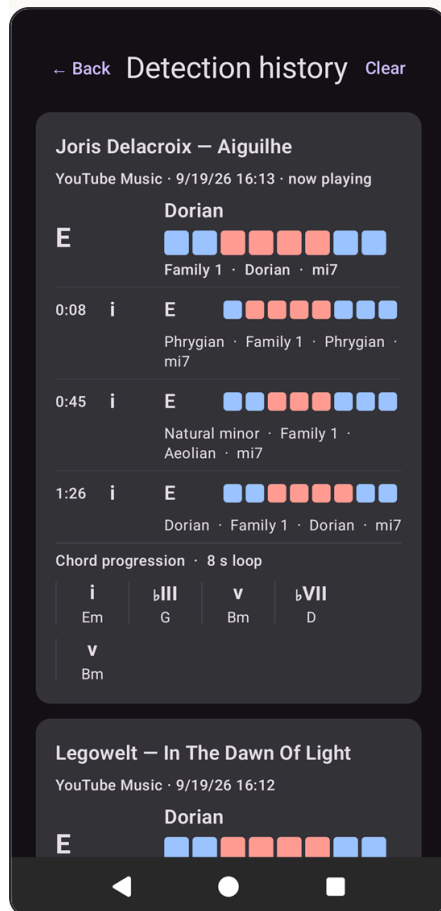
Degrees are read on the **major scale of the tonic**, with $\flat$ or $\sharp$ for other pitches: in A minor, F major is written $\flat$ VI. Upper case is major (+ for augmented), lower case is minor ($^\circ$ for diminished).

Analyse again

The **Analyse again** button forgets everything and analyses the current piece afresh — useful if you started listening in the middle of an unusual passage.

The reading is deliberately **slow and steady**. The first estimate needs about 8 seconds of music; a change of scale is only accepted if it lasts 12 seconds, a change of mode 30 seconds. Expect the first minute of a piece to be the least reliable.

6 • History, one section per piece



One section per piece: main scale, scales held in turn with their degree, chord progression.

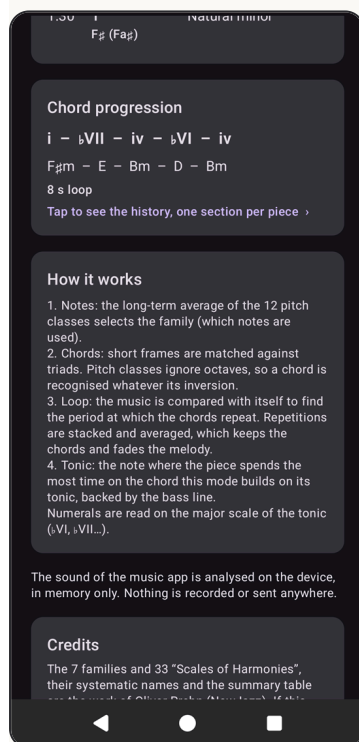
Tap the **Chord progression** card of the Harmony panel to open the history. The piece being played comes first, marked *now playing*; earlier pieces follow, most recent first, up to fifty.

Each section reads in three levels, separated by thin lines:

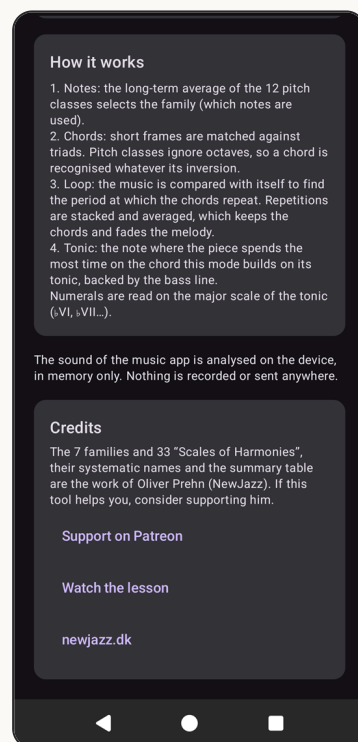
- **The main scale** of the piece, under its name: the key in large letters, the scale and its **colour tiles**.
- **The scales held in turn**, one row each, in fixed columns: the time, the **degree in Roman numerals** relative to the main key, the tonic, the tiles. Read down a column to follow the move from one degree to the next.
- **The chord progression**, at the foot: one cell per chord, the degree above its name, and the length of the loop.

Pieces are told apart by the title and artist that the music app announces itself — the same information your lock screen shows. Nothing is recognised by listening and nothing is looked up online. When an app announces no title, a silence of a few seconds marks the boundary and the section is called *Unknown piece*.

Clear empties the list. The history is a private file of the app; it contains names of pieces, scales and chords, never any sound.



The panel also explains the method in a few lines, and states what happens to the sound.



Credits, with links to Oliver Prehn's lesson, website and Patreon page.

7 • The Scales of Harmonies sheet

The link  **The 33 scales, colour-coded (PDF)**, in the Harmony card and in the panel, opens a one-page study sheet inside the app. Pinch to zoom, drag to move around. It exists in English, French and Spanish and follows the app's language.

The Scales of Harmonies

An overview of 33 scales in 7 families, by **Oliver Prehn** (NewJazz) — colour-coded study sheet

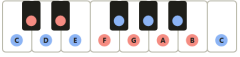
The 33 scales presented in the NewJazz lesson are organised in 7 families. Each family has one interval pattern; a scale is that pattern read from a given degree — which is why the rows of a family line up as a staircase. The 33 scales follow from a few simple rules:

- Only half, whole and whole-and-a-half steps
- Half steps may not be neighbours
- Whole and whole-and-a-half steps may not be neighbours
- Whole-and-a-half steps may not be neighbours

Following them limits the total to 33 scales, all well suited to building clear, evident harmonies.

Lesson: youtu.be/Vq2xt2D3e3E - www.newjazz.dk - Support Oliver Prehn: patreon.com/newjazz - paypal.me/newjazz

Reading the colours



• Blue and • red are the two whole-tone sets of the keyboard. A **whole step** (1) stays in the same colour; a **half step** (1/2) or a **step and a half** (1 1/2) crosses to the other. Each scale is a row of tiles, one per note, from the tonic to its octave; that two-colour sequence is the fingerprint of the mode. Colours are shown with the first scale of each family starting on C; down a family, each column is one and the same note. From a red tonic, swap the two colours.

FAMILY	POPULAR NAME	INTERVAL STEPS		*SYSTEMATIC NAME	CHORDS
1	Major	1-1-½-1-1-1-½		Ionian	Ma7
	Dorian	1-½-1-1-1-½-1		Dorian	mi7
	Phrygian	½-1-1-1-½-1-1		Phrygian	mi7
	Lydian	1-1-1-½-1-1-½		Lydian	Ma7
	Mixolydian	1-1-½-1-1-½-1		Mixolydian	7
	Natural minor	1-½-1-1-½-1-1		Aeolian	mi7
	Locrian	½-1-1-½-1-1-1		Locrian	mi7b5
2	Altered / Super Locrian	½-1-½-1-1-1-1		Ionian #1	7alt / mi7b5
	Ascending mel. minor	1-½-1-1-1-1-½		Dorian #7	miMa7
	Dorian b2	½-1-1-1-1-½-1		Phrygian #6	mi7
	Lydian Augmented	1-1-1-1-½-1-½		Lydian #5	Ma7#5
	Lydian dominant	1-1-1-½-1-½-1		Mixolydian #4	7
	Aeolian dominant	1-1-½-1-½-1-1		Aeolian #3	7
	Half diminished	1-½-1-½-1-1-1		Locrian #2	mi7b5
3	Major #5 / Major Aug.	1-1-½-1½-½-1-½		Ionian #5	Ma7#5
	Dorian #4	1-½-1½-½-1-½-1		Dorian #4	mi7
	Phrygian dominant	½-1½-½-1-½-1-1		Phrygian #3	7
	Lydian #2	1½-½-1-½-1-1-½		Lydian #2	Ma7
	Altered dominant bb7	½-1-½-1-1-½-1½		Mixolydian #1	dim7
	Harmonic minor	1-½-1-1-½-1½-½		Aeolian #7	miMa7
	Locrian #6	½-1-1-½-1½-½-1		Locrian #6	mi7b5
4	Harmonic Major	1-1-½-1-½-1½-½		Ionian b6	Ma7
	Dorian b5	1-½-1-½-1½-½-1		Dorian b5	mi7b5
	Phrygian b4	½-1-½-1½-½-1-1		Phrygian b4	mi7 / 7
	Lydian b3	1-½-1-½-1½-½-1		Lydian b3	miMa7
	Mixolydian b2	½-1½-½-1-1-½-1		Mixolydian b2	7
	Lydian augmented #2	1½-½-1-1-½-1-½		Aeolian b1	Ma7#5 / dim7
	Locrian bb7	½-1-1-½-1-½-1½		Locrian b7	dim7
5	Diminished	1-½-1-½-1-½-1-½		Diminished	dim7
	Dominant diminished	½-1-1-1-1-1-1		Inverted diminished	7
6	Whole tone	1-1-1-1-1-1		Whole tone	7 #5/b5
	Augmented	1½-½-1½-½-1½-½		Augmented	Ma7
7	Inverted Augmented	½-1½-½-1½-½-1½		Inverted Augmented	6#5

*The systematic names do not follow the jazz-school habit of naming every scale after the Major scale. *Aeolian #7 is an Aeolian scale with a raised 7th step: Aeolian is the premise, #7 the alteration. The first 4 families thus run from Ionian to Locrian with a simple sequence of sharps or flats.

Musical content © Oliver Prehn, reproduced for study with credit; all rights remain with its author. Layout and colour code: Steve Nodlock, DuoMix Fader, 2026. Handwritten notes on the source sheet are not included.

The sheet reproduces the data of Oliver Prehn's overview — 33 scales in 7 families, with popular name, interval steps, systematic name and chords — and adds a colour code:

-  **Blue** and  **red** are the two whole-tone sets of the keyboard: C D E F# G# A# and C# D# F G A B.
- Each scale is a **row of square tiles**, one per note, from the tonic to its octave — the same notation as the notification card and the history. A **whole step** stays in the same colour; a **half step** or a **step and a half** crosses to the other.
- Colours are given with the first scale of each family starting on C. The scales of a family line up as a **staircase** — each one is the family's pattern read from the next degree — so every column is one and the same note. From a red tonic (C#, F, G, A...), swap the two colours: the app always shows the real ones.
- Each family has its own colour; the scales printed in bold in the original are bold here too.

The musical content of this sheet is Oliver Prehn's work and remains his. Only the layout and the colour code were added. See *Credits*.

8 • How the analysis works, and its limits

1. **Notes** → **family**. The energy of the 12 pitch classes is averaged over about thirty seconds. The set of notes that explains it best gives the family. When two candidates are close, the more common family wins.
2. **Chords**. Short frames are matched against the 48 triads. Pitch classes ignore octaves, so a chord is recognised whatever its inversion or voicing.
3. **Loop**. The music is compared with itself to find the period at which the chords come back. All repetitions are stacked and averaged: what returns every time — the chords — is reinforced; melody, voice and drums fade. If no loop is found, the last 20 seconds are read as they are, and the panel says so.
4. **Tonic** → **mode**. The modes of a family share the same notes, so notes alone cannot tell D Dorian from C major. The tonic is the note where the piece spends the most time on the chord that this mode builds on its first degree — the “Chords” column of the table — backed by the bass line.

Limits you should know

- **The mode is less certain than the family**, and sometimes the ambiguity is musical: a loop such as I – V – vi – IV spends as much time on four chords, each of which is the tonic chord of a mode of the same family.
- Rootless jazz voicings can hide a chord (a rootless Cmaj9 is an Em7); the bass line is then the only clue.
- Chords shorter than about half a second, music tuned far from A = 440 Hz, and very percussive mixes are read poorly. Four-note chords are reduced to their triad.
- If the music channel is faded to zero, there is nothing left to analyse.
- The method has been tested on synthetic signals and on a handful of real pieces, not on an annotated corpus: its success rate on real music is unknown.

The full method, its constants and what did not work are described in the research note shipped with the source code, `docs/research/harmony-analysis.md`.

9 • Privacy

- The app has **no network permission**. Nothing it hears, reads or stores can leave the phone.
- To analyse a piece it listens to the **music app only**. The sound is reduced on the spot to twelve numbers per instant and discarded; it is never recorded.
- It reads the title and artist that the music app publishes for the lock screen, to name the sections of the history.
- The history is a private file; **Clear** deletes it. Uninstalling the app removes everything.
- The privileged service it starts through Shizuku can only do a short, fixed list of things — change the audio-focus setting and the volume of the listed apps, listen to the music app, read its title, open DuoMix Fader's own screen. It runs no arbitrary command and stops by itself when the app stops.

10 • Credits

The 7 families and 33 *Scales of Harmonies*, their systematic names and the overview table are the work of **Oliver Prehn** (NewJazz). If this tool helps you, please support him.

- Lesson: [ALL ABOUT MUSICAL SCALES – A COMPLETE GUIDE!!!](https://youtu.be/Vq2xt2D3e3E) — youtu.be/Vq2xt2D3e3E
- Website: www.newjazz.dk · Support: patreon.com/newjazz · paypal.me/newjazz

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